

WorkNow

Student Gig Marketplace Platform

Entity-Relationship (ER) Schema Specification

Database Tables & Data Dictionary Documentation

TARGET ENVIRONMENT: MySQL 8.0+

Version: 1.0.0 (Production Grade)

Author: Antigravity AI Code Architect

1. Schema Architecture & Overview

The WorkNow database schema is designed with structural isolation, normalized relationships, and strict referential integrity constraint rules matching standard transactional platforms. It consists of six main tables supporting authentication, profiles, verification, job matching, escrows, wallet movements, and rating collections.

Entity Relationships List:

Relation Keys	Type	Description / Scope
users.id -> verification_documents.user_id	One-to-Many	Tracks worker verification files
users.id -> jobs.employer_id	One-to-Many	Audits jobs posted by a customer
users.id -> jobs.worker_id	One-to-Many	Tracks hired student worker per gig
users.id -> applications.worker_id	One-to-Many	Tracks worker bids to active jobs
jobs.id -> applications.job_id	One-to-Many	Housed applications list for a gig
users.id -> transactions.user_id	One-to-Many	Wallet transaction ledger records
jobs.id -> reviews.job_id	One-to-One	Linkage between completed task & rating
users.id -> reviews.reviewer_id / reviewee_id	One-to-Many	Identifies sender/receiver of feedback

Mermaid ER Diagram Source Code:

```
erDiagram
    users ||--o{ verification_documents : "uploads"
    users ||--o{ jobs : "posts (employer)"
    users ||--o{ jobs : "accepts (worker)"
    users ||--o{ applications : "submits"
    users ||--o{ transactions : "performs"
    users ||--o{ reviews : "receives/writes"
    jobs ||--o{ applications : "has"
    jobs ||--o{ reviews : "linked_to"
```

2. Data Dictionary Specifications

Table: users

Column Name	Type	Attributes	Description
id	INT	PK, AUTO_INC	Primary key identifier
phone	VARCHAR(20)	UNIQUE, NULLABLE	Mobile number (Auth key)
password_hash	VARCHAR(255)	NULLABLE	Salted password hash value
auth_provider	VARCHAR(50)	DEFAULT 'PHONE'	PHONE EMAIL GOOGLE GITHUB
provider_id	VARCHAR(255)	NULLABLE	Social provider user ID
role	VARCHAR(20)	DEFAULT 'WORKER'	WORKER CUSTOMER ADMIN
name	VARCHAR(100)	NULLABLE	Display name
email	VARCHAR(100)	NULLABLE	Email address
verification_status	VARCHAR(20)	DEFAULT 'PENDING'	PENDING VERIFIED REJECTED
rating	FLOAT	DEFAULT 0.0	Average peer rating score
total_ratings	INT	DEFAULT 0	Count of reviews submitted
skills	TEXT	NULLABLE	Comma separated skills tag list
wallet_balance	DECIMAL(10,2)	DEFAULT 0.00	Digital wallet cache funds
is_online	BOOLEAN	DEFAULT FALSE	Presence tracking status

Table: verification_documents

Column Name	Type	Attributes	Description
id	INT	PK, AUTO_INC	Primary key identifier
user_id	INT	FK, NOT NULL	References users(id)
type	VARCHAR(50)	NOT NULL	STUDENT_ID AADHAR PAN SELFIE
url	VARCHAR(500)	NOT NULL	Mock upload file path
status	VARCHAR(20)	DEFAULT 'PENDING'	PENDING VERIFIED REJECTED

Table: jobs

Column Name	Type	Attributes	Description
id	INT	PK, AUTO_INC	Primary key identifier
employer_id	INT	FK, NOT NULL	References users(id) (Employer)
worker_id	INT	FK, NULLABLE	References users(id) (Hired Worker)
title	VARCHAR(150)	NOT NULL	Title of the gig task
description	TEXT	NOT NULL	Scope details of the gig
category	VARCHAR(50)	NOT NULL	Delivery Cleaning Helper Tutoring...
budget	DECIMAL(10,2)	NOT NULL	Locked budget escrow amount
location_lat	DOUBLE	NOT NULL	Task latitude coordinate
location_lng	DOUBLE	NOT NULL	Task longitude coordinate
status	VARCHAR(25)	DEFAULT 'OPEN'	OPEN ASSIGNED IN_PROGRESS COMPLETED

Table: applications

Column Name	Type	Attributes	Description
id	INT	PK, AUTO_INC	Primary key identifier
job_id	INT	FK, NOT NULL	References jobs(id)
worker_id	INT	FK, NOT NULL	References users(id) (Worker)
status	VARCHAR(20)	DEFAULT 'PENDING'	PENDING HIRED REJECTED
cover_letter	TEXT	NULLABLE	Worker proposal message details

Table: transactions

Column Name	Type	Attributes	Description
id	INT	PK, AUTO_INC	Primary key identifier
user_id	INT	FK, NOT NULL	References users(id)
amount	DECIMAL(10,2)	NOT NULL	Transactional money movement
type	VARCHAR(20)	NOT NULL	CREDIT DEBIT
description	VARCHAR(255)	NOT NULL	Transaction description notes

Table: reviews

Column Name	Type	Attributes	Description
id	INT	PK, AUTO_INC	Primary key identifier
job_id	INT	FK, NOT NULL	References jobs(id)
reviewer_id	INT	FK, NOT NULL	References users(id) (Author)
reviewee_id	INT	FK, NOT NULL	References users(id) (Target)
rating	INT	NOT NULL	Stars rating (1 to 5)
comment	TEXT	NULLABLE	Feedback commentary text